Embedded Reinforcement Learning

Reference	:	HAE1
Supervisor	:	Prof Herman Engelbrecht
Can be completed in first semester	:	No
Students allowed for this topic	:	1
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Reinforcement learning (RL) is a field of machine learning concerned with agents learning how to take actions in an environment in order to maximise a cumulative reward. Deep Reinforcement Learning (DRL) combines deep learning and reinforcent learning to solve complex tasks.
The purpose of this topic is to implement and train an embedded RL agent to solve scenarios in the Miniworld (https://miniworld.farama.org/) virtual environment (an environment simulator for reinforcement learning and robotics research supported by the Farama Foundation). The topic will first involve implementing and training an DRL agent on desktop computer using images obtained from the virtual environment. The purpose is to train a policy network and obtain the baseline performance on a desktop computer. After succesfully training and evaluating an DRL agent, the next step is to compress the policy network so that it can be deployed onto an embedded microcontroller unit (MCU) such as the Raspberry Pi 2 Pico W. An additional challenge is to use a Raspberry Pi camera to obtain the images from a computer screen.
A solid foundation in programming and mathematics is needed for this project. The student will gain experience in reinforcement learning, deep learning and implementing algorithms from these fields on MCUs using Python, C/C++ and the PyTorch deep learning framework.

Key modules (Scientific and Engineering Knowledge):
Computer Systems (214) | Computer Systems (245) | Design (E) (314)